

IT Infrastructure

Agenda – Day 1

- Introduction to Networking Devices
- Overview of LAN and WAN technologies
- IPv4 and IPv6
- Datacenter Overview
- Server form factor
- Server Management portal (IPMI/iLO, iDrac)
- Server Events
- Basics of storage (NAS, DAS and SAN)
- Comparing Fibre Channel, iSCSI, and Fibre Channel over Ethernet



Agenda – Day 2

- Concept of RAID
- Introduction to Operating systems (Server and Client)
- Concept of Domain and Workgroup
- File Systems (FAT, NTFS, ReFS, EXT3, EXT4 etc)
- File system management
- Installing Operating system (Windows 10, Server 2016, Linux
- Installing OS using management Console
- Firewall (Hardware and Software)



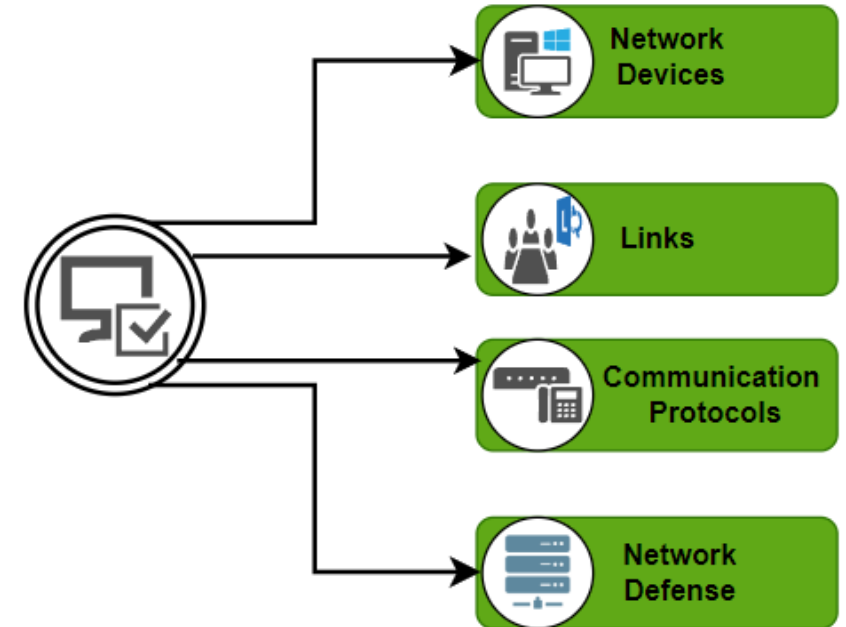
What is a Computer Network?

- A computer network is a system that connects many independent computers to share information (data) and resources.
- The integration of computers and other different devices allows users to communicate more easily.
- A network connection can be established using either cable or wireless media.
- Hardware and software are used to connect computers and tools in any network.



Key Components of a Computer Network

- A computer network is made up of two main parts:
 - **Devices** (called nodes) &
 - **Connections** (called links)
- The links connect the devices to each other.
- The rules for how these connections send information are called communication protocols.
- The starting and ending points of these communications are often called ports.



Port Numbers

- In networking, ports are categorized primarily based on their port numbers, which range from **0 to 65535**.
- These ports are divided into three main categories as defined by the Internet Assigned Numbers Authority (**IANA**):
 - **Well-Known Ports** (0 – 1023)
 - **Registered Ports** (1024 – 49151)
 - Dynamic/Private/**Ephemeral Ports** (49152 – 65535)



Port Numbers – Well-Known Ports

- **Ranges** from 0 – 1023
- **Used by:** Common system or well-known services.
- **Reserved for:** Privileged services and processes.
- **Examples:**
 - 20, 21 – FTP (File Transfer Protocol)
 - 22 – SSH (Secure Shell)
 - 23 – Telnet
 - 25 – SMTP (Simple Mail Transfer Protocol)
 - 53 – DNS (Domain Name System)
 - 80 – HTTP (HyperText Transfer Protocol)
 - 443 – HTTPS (HTTP Secure)



Port Numbers – Registered Ports

- **Ranges** from 1024 – 49151
- **Used by:** User processes or applications.
- Registered with IANA for specific services by software developers.
- Examples:
 - 1433 – Microsoft SQL Server
 - 3306 – MySQL Database
 - 3389 – Microsoft RDP (Remote Desktop Protocol)
 - 8080 – HTTP alternative port (often used by web servers)



Port Numbers – Dynamic/Private/Ephemeral Ports

- **Ranges** from 49152 – 65535
- **Used by:** Temporary connections, usually client-side.
- Assigned dynamically by the operating system when an application requests a network connection.
- **Examples:**
 - Any port in this range used for NAT (Network Address Translation), temporary communication, or client-side connections in TCP/UDP communications.



Networking devices

- Networking devices are hardware components used to connect computers, printers, phones, cameras, servers, and other devices to a network, enabling them to communicate and share data.
- These devices perform functions like:
 - *Routing data* to the correct destination
 - *Filtering or managing* traffic
 - *Connecting* different types of networks



Common Networking Devices

- *Router*
- *Switch*
- *Hub*
- *Modem*
- *Access Point*
- *Firewall*
- *Gateway*
- *Network Interface Card (NIC)*
- *Repeater*
- *Bridge*



Router

- Connects **multiple networks** (e.g., LAN to the Internet).
- Routes data packets based on **IP addresses**.
- Determines **best path for data** to reach its destination.
- Often includes **Wi-Fi functionality** in home routers.
- Offers basic security features like **firewall** and **NAT** (Network Address Translation).
- Supports **DHCP** to assign IP addresses to devices.
- Has **WAN** and **LAN** ports for external and internal network connections.
- Maintains a **routing table** to manage data flow.
- Can be configured via **web interface** or **CLI**.



Switch

- Connects multiple devices within the **same LAN**.
- Forwards data only to the **specific device** (MAC address) that needs it.
- Maintains a **MAC address table** to map ports to devices.
- Operates at **Layer 2** (Data Link Layer) of the OSI model (some at Layer 3 with routing capability).
- Provides **faster and more efficient data** transmission than a hub.
- Has **multiple Ethernet ports** (e.g., 8, 24, 48 ports).
- Used in **offices, data centers, and enterprise** networks to interconnect devices.
- May be managed or unmanaged:
 - **Managed:** configurable, supports VLANs, monitoring, etc.
 - **Unmanaged:** plug-and-play, no configuration needed.
- Reduces network collisions and **improves performance** over hubs.



Hub

- Connects **multiple devices in a LAN**.
- **Broadcasts incoming data** to all ports, regardless of destination.
- **Does not store MAC** addresses or make intelligent forwarding decisions.
- Operates at **Layer 1 (Physical Layer)** of the OSI model.
- Slower and less efficient compared to switches.
- Typically has **multiple Ethernet ports** (e.g., 4, 8, 16).
- **Unmanaged device** – no configuration required.
- **Rarely used today**, mostly replaced by switches.
- Useful in basic setups or labs for understanding network behavior.



Modem

- Modem = **Modulator + Demodulator**.
- Converts **digital signals** from a computer to **analog signals** for transmission over phone/cable lines.
- Connects a **home** or **office** network to the Internet via an ISP.
- Typically has one port for the ISP line and one for connecting to a router or computer.
- Works at the **Physical Layer (Layer 1)** of the OSI model.
- **Does not manage internal network** traffic – only handles the internet connection.
- **Required for internet** access in homes and small businesses.



Access Point

- Provides **wireless connectivity** to devices within a network.
- Connects to a wired network (e.g., via Ethernet) and **creates a Wi-Fi network**.
- **Acts as a bridge** between wireless clients and wired LAN.
- **Extends Wi-Fi** coverage in large areas (offices, campuses, hotels).
- Operates at **Layer 2** (Data Link Layer) of the OSI model.
- Supports **multiple devices** simultaneously (smartphones, laptops, etc.).
- Not the same as a router — it **doesn't route traffic** between networks.
- Used to **improve scalability, range, and performance** of wireless networks.



Firewall

- **Monitors and controls** incoming and outgoing network traffic.
- Acts as a **barrier** between trusted and untrusted networks (e.g., LAN and Internet).
- Uses **predefined rules** to allow or block data packets.
- Filters traffic based on **IP address, port number, protocol, or content**.
- Operates primarily at **Layer 3 (Network)** and **Layer 4 (Transport)** of the OSI model.
- Protects against **unauthorized access, malware, and attacks**.
- **Configurable by administrators** for custom security policies.



Gateway

- Connects two different networks that use different protocols.
- Acts as a translator or protocol converter between dissimilar systems.
- Enables communication between LAN and WAN, or enterprise network and internet.
- Operates at all seven layers of the OSI model (depending on implementation).
- *Commonly used to connect internal networks to external networks (e.g., the Internet).*
- Can include routing, firewall, and NAT functionality.
- Can be implemented as hardware, software network as an entry/exit point, or cloud-based service.
- In home/office networks, the router often acts as the default gateway.



Network Interface Card (NIC)

- Enables a device (computer, printer, server, etc.) to connect to a network.
- Provides a physical interface (e.g., Ethernet port or wireless adapter).
- Can be wired (Ethernet NIC) or wireless (Wi-Fi NIC).
- Has a unique MAC (Media Access Control) address for identification.
- Operates at Layer 2 (Data Link Layer) and interfaces with Layer 1 (Physical Layer).
- Can be integrated into the motherboard or added as a separate expansion card.
- Essential for network communication in desktops, laptops, and servers.
- Transmits and receives data packets over the network.
- Supports various speeds like 10/100/1000 Mbps or 10 Gbps and beyond.
- Required for both LAN and Internet access.



Repeater

- Amplifies and retransmits network signals to extend the range.
- Used to boost weak or degraded signals in a network.
- Helps maintain signal strength over long distances.
- Connects two network segments using the same protocol.
- Operates at Layer 1 (Physical Layer) of the OSI model.
- Has no filtering or addressing capabilities — just regenerates signals.
- Common in wired (Ethernet) and wireless (Wi-Fi extender) networks.
- Useful when cabling exceeds the standard-length limit (e.g., 100 meters for Ethernet).
- Does not reduce network traffic or collisions — just extends coverage.
- Ideal for simple network expansion in homes or small offices.



Bridge

- Connects and filters traffic between two or more LAN segments.
- Forwards data based on MAC addresses.
- Maintains a MAC address table to determine which segment to forward traffic to.
- Helps reduce network traffic and collisions by dividing large networks.
- Operates at Layer 2 (Data Link Layer) of the OSI model.
- Connects networks using the same protocol (unlike a gateway).
- Can segment a busy network to improve performance and efficiency.
- Typically has two or more ports for connecting network segments.
- Useful in legacy or small networks before switches became common.
- Can be implemented as hardware or software (e.g., in virtual environments).



Server

- A server is a computer or system that provides services, resources, or data to other computers (clients) over a network.
- Servers are designed to handle multiple requests and provide consistent, centralized functionality like file storage, websites, email, databases, or applications.
- Examples:
 - Web server,
 - File server,
 - Mail server,
 - Database server.



Client

- A client is a computer or software that requests services or resources from a server.
- The client initiates the communication with the server and waits for a response.
- Examples:
 - Web browser,
 - email app, or
 - any software accessing a remote system.



Difference Between Server and Client

Feature	Server	Client
Role	Provides services or resources	Requests services or resources
Initiates Communication	Usually responds to client requests	Initiates connection to the server
Hardware Requirements	Usually more powerful, runs 24/7	Can be less powerful
Examples	Web server, Mail server, FTP server	Chrome browser, Outlook email client
Location	Centralized (data center or cloud)	Can be anywhere (user device)
Dependency	Operates independently or serves many clients	Depends on server for services



Out-of-Band Management Tools

- They are part of server management hardware used for remote management and monitoring of servers independent of the operating system or server power state.

iLO (HP)	Out-of-Band Management	HP's proprietary remote server management solution
iDRAC (Dell)	Out-of-Band Management	Dell's proprietary tool for remote management of PowerEdge servers
IPMI (Standard)	Out-of-Band Management Protocol	Open industry standard for remote server management, supported by many vendors



What Out-of-Band Management Tools Do?

- Power on/off or reboot servers remotely
- View hardware health (temperature, fan speed, power status)
- Access the server console (even if OS is down)
- Mount ISO images for remote OS installation
- Manage server BIOS/firmware settings



Server Form Factor

- Server Form Factor refers to the physical shape, size, and design of a server system.
- It defines how a server is built, how it's deployed, and where it fits—whether in a rack, on a desk, or in a blade enclosure.

Form Factor	Description	Use Case
Rack Server	Slim, rectangular servers mounted in a standard 19-inch rack (1U, 2U, etc.)	Data centers, enterprises
Tower Server	Looks like a traditional desktop computer tower	Small offices, branch locations
Blade Server	Very thin servers (blades) that slide into a blade chassis	High-density computing environments
Modular Server	Combines features of blade and rack; customizable modules in one chassis	Scalable environments
Microserver	Small, energy-efficient server for lightweight workloads	Web hosting, development labs



Server events

- These are system generated notifications or logs
- These logs helps admins in:
 - Monitor
 - Troubleshooting
 - Maintenance.



Types of Server events:

- *Hardware events* generated by CPU, RAM, HDD, Power supply
- *Power events* unexpected power off, failures
- *Thermal events* overheating warnings, temperatures
- *Boot/POST events* events got generated at boot/POST level.
- *OS events* OS reboots, kernel panic, service startup or failure.
- *Security events* login attempts, breaches
- *Network events* link up/down, IP conflict, NIC failure
- *Storage events* disk failures
- *IPMI events* remote access activities, sensor issues.



Basics of Storage

- Storage in computing refers to the technology used to save, retain, and retrieve data.
- It is a fundamental part of any computer system, server, or data center.
- Types of Storage (by Functionality)

Type	Description	Examples
Primary Storage	Directly accessed by the CPU; fast but volatile	RAM, Cache Memory
Secondary Storage	Persistent data storage for OS, apps, files	HDDs, SSDs
Tertiary Storage	Used for backups or long-term archiving	Tape Drives, Optical Disks
Offline Storage	Manually connected storage for portability	USB Drives, External HDDs



Types of Storage Devices

Type	Characteristics	Use Case
HDD (Hard Disk Drive)	Spinning disks, slower, cheaper, large capacity	File servers, backups
SSD (Solid State Drive)	Faster, no moving parts, more expensive	OS boot, applications, VMs
NVMe SSD	Very fast, uses PCIe interface	High-performance workloads
Optical (CD/DVD)	Older tech, slow, used for backups	Archiving, media
Tape Storage	Very large capacity, used for backup/archive	Cold storage, disaster recovery



Key Storage Concepts

Concept	Description
Capacity	Amount of data that can be stored (GB, TB, PB)
Read/Write Speed	How fast data is accessed or saved
Latency	Delay before data transfer begins
IOPS (I/O per second)	Number of input/output operations per second
Throughput	Amount of data transferred over time



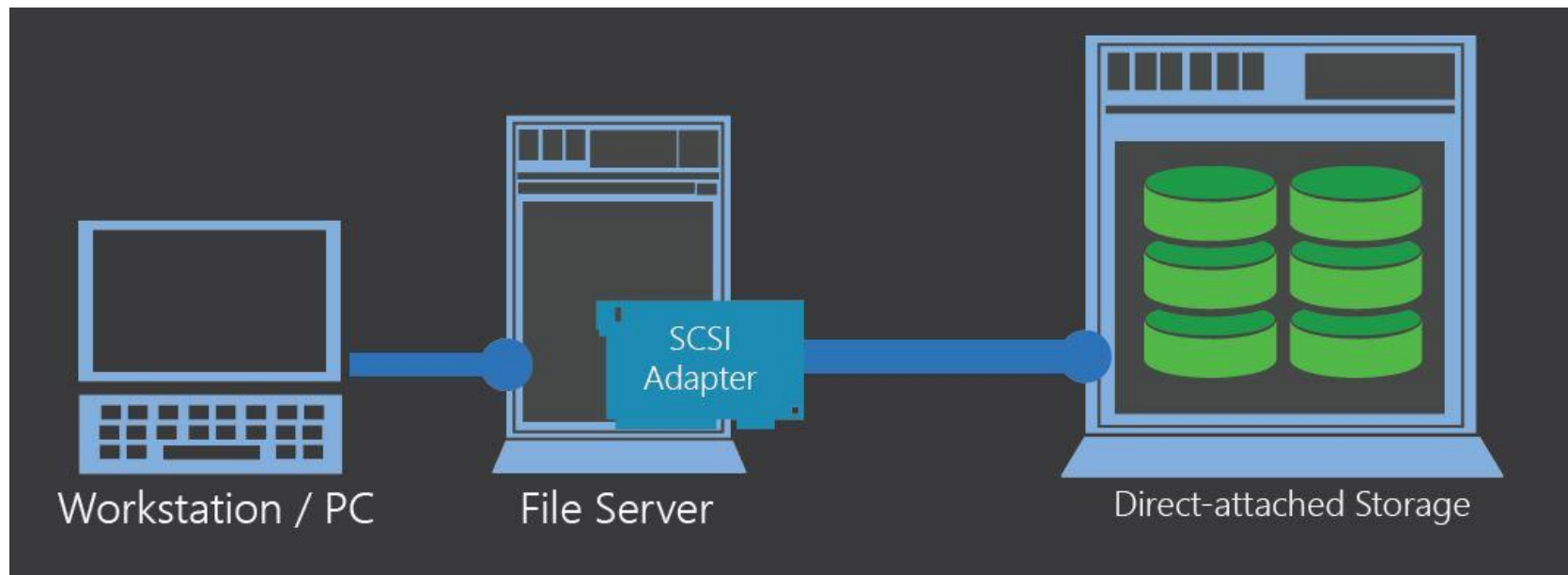
Storage Architecture Types

Type	Description
DAS (Direct Attached Storage)	Directly connected to one computer (e.g., internal disk)
NAS (Network Attached Storage)	Shared storage over network; file-level access
SAN (Storage Area Network)	High-speed network for block-level storage sharing
Cloud Storage	Internet-based storage from providers like AWS, Azure



Direct Attached Storage

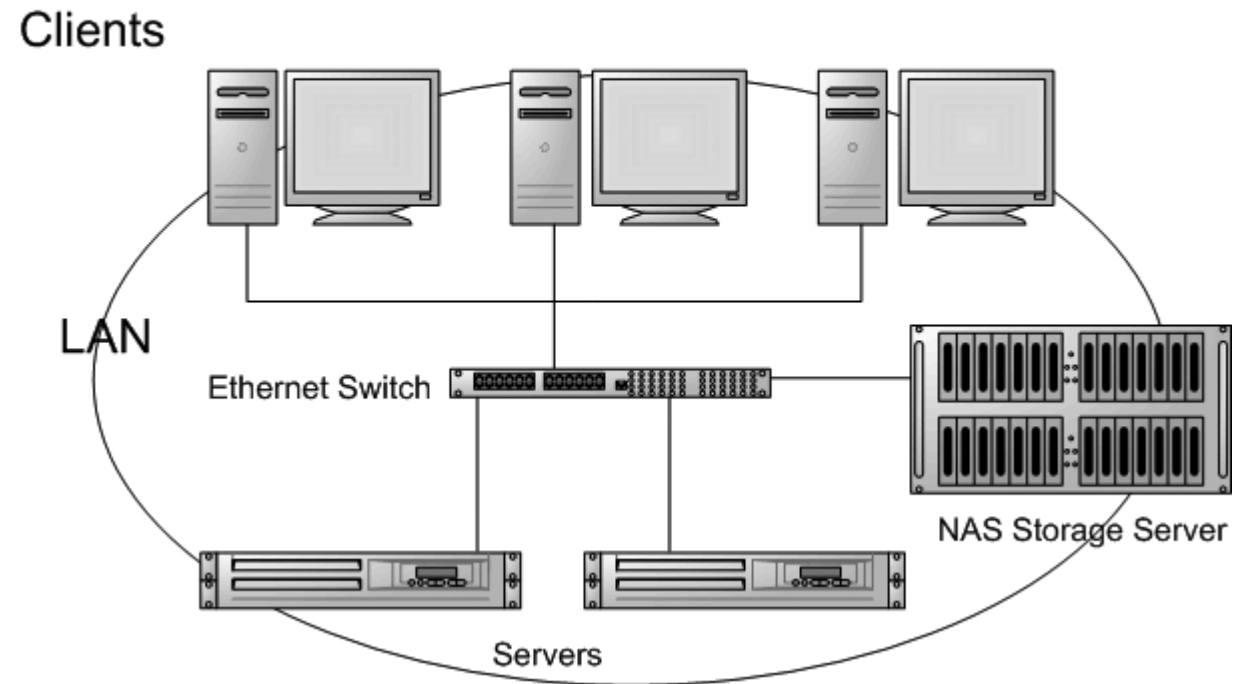
- **DAS** is a storage system that is directly connected to a computer or server without using a network.
- It provides a dedicated storage solution for a specific server or workstation, making it a simple and cost-effective option for data storage needs.



Network Attached Storage

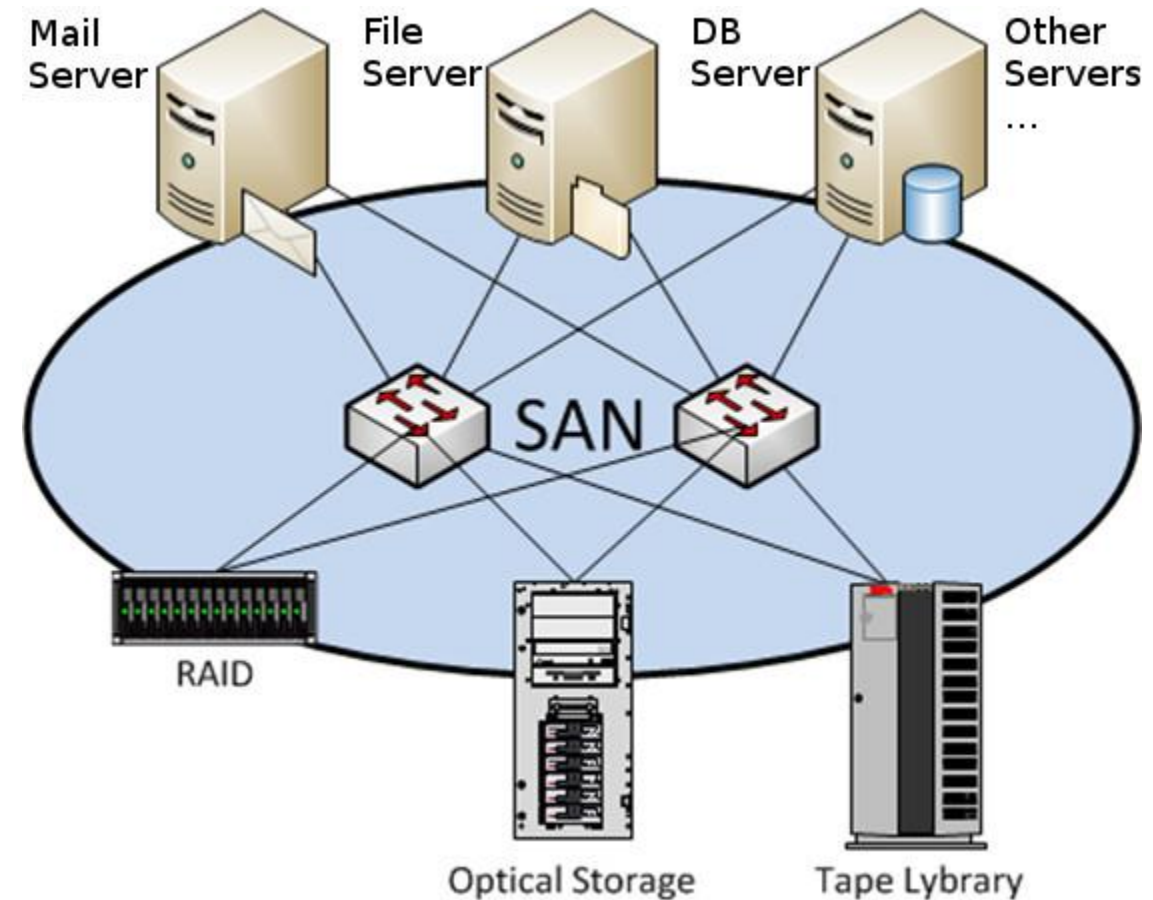
- **NAS** is a data storage device connected to a local network that allows multiple users or devices to access and share files stored on it.
- It provides centralized, scalable storage that can be accessed remotely over a network, making it convenient for home or small office data backup and file sharing.

Network Attached Storage



Storage Area Network

- A **SAN** is a dedicated high-speed network that provides access to consolidated, block-level data storage.
- It allows multiple servers to access shared storage devices, such as disk arrays and tape libraries, as if they were locally attached.
- This enables efficient data management, backup, and disaster recovery across an organization's storage infrastructure.



Server Operating System

- The job is to serve to the client
- Server serves the request based on the need.
- Ex:
 - DHCP server
 - DNS server
- List of Windows servers
 - Windows Server NT
 - Windows Server 2003
 - Windows Server 2008
 - Windows Server 2008R2
 - Windows Server 2012
 - Windows Server 2012R2
 - Windows Server 2016
 - Windows Server 2019
 - Windows Server 2022
 - Windows Server 2025



Client Operating System

- The job is to request services from server
- Client has very limited permission
- List of Windows Client OS:
 - Windows XP
 - Windows Vista
 - Windows 7
 - Windows 8.0
 - Windows 8.1
 - Windows 10
 - Windows 11



Assign IP address to server/client OS:

Static IP

Internet Protocol Version 4 (TCP/IPv4) Properties

General

You can get IP settings assigned automatically if your network supports this capability. Otherwise, you need to ask your network administrator for the appropriate IP settings.

☐ Obtain an IP address automatically

☒ Use the following IP address:

IP address: 192 . 168 . 10 . 10

Subnet mask: 255 . 255 . 255 . 0

Default gateway: 192 . 168 . 10 . 1

☐ Obtain DNS server address automatically

☒ Use the following DNS server addresses:

Preferred DNS server: 192 . 168 . 10 . 10

Alternate DNS server: 8 . 8 . 8 . 8

☐ Validate settings upon exit

Advanced...

OK Cancel

Automatic IP

Internet Protocol Version 4 (TCP/IPv4) Properties

General Alternate Configuration

You can get IP settings assigned automatically if your network supports this capability. Otherwise, you need to ask your network administrator for the appropriate IP settings.

☒ Obtain an IP address automatically

☐ Use the following IP address:

IP address: . . .

Subnet mask: . . .

Default gateway: . . .

☒ Obtain DNS server address automatically

☐ Use the following DNS server addresses:

Preferred DNS server: . . .

Alternate DNS server: . . .

☐ Validate settings upon exit

Advanced...

OK Cancel

APIPA

```
Jeetu> Get-NetIPAddress | Format-Table
```

ifIndex	IPAddress
8	fe80::daac:109f:4442:a7c9%8
5	fe80::50fe:e82a:ee1b:5237%5
3	fe80::a3d0:b24f:f01d:f046%3
6	fe80::a3:87f4:d40e:ae69%6
10	fe80::2c65:24e7:ef29:2bd%10
13	fe80::d114:25df:82b2:867f%13
13	2401:4900:1ca2:4739:44b1:e136:56d9:c054
13	2401:4900:1ca2:4739:37f7:82ee:646d:e891
1	::1
8	192.168.88.1
5	192.168.245.1
3	169.254.0.204
6	169.254.53.147
10	169.254.162.54
13	192.168.1.102
1	127.0.0.1



File System

- A file system controls how data is stored and retrieved.
- Without a file system, information placed in a storage device would be one large block of data with no way to tell where one piece of data ends and the next begins.
- Responsibilities of a File System
 - Organizing files and directories
 - Tracking location of data on disk
 - Managing access permissions (read/write/execute)
 - Ensuring data integrity and consistency
 - Handling metadata (file size, timestamp, owner, etc.)



Types of File Systems

File System	Platform	Key Features
FAT32	Windows, USB	Old, limited to 4GB files, widely supported
NTFS	Windows	Journaling, permissions, encryption, large file support
exFAT	Windows, Mac	Modern FAT version; supports large files and USBs
ext3	Linux	Journaling, stable
ext4	Linux	Improved ext3, supports large volumes and files
XFS	Linux	High performance, journaling, good for large files
Btrfs	Linux	Snapshot, pooling, copy-on-write (modern filesystem)
ZFS	Solaris, FreeBSD	Pooled storage, checksums, high data integrity
HFS+/APFS	macOS	Apple file systems, APFS is newer and faster
ISO9660	CD/DVD	Standard for optical media



Storage categories

- MBR
- GPT
- Basic disks
- Dynamic disks
- Storage spaces
- RAID
- iSCSI



MBR – Master Boot Record

- MBR is a partitioning scheme used in storage devices (like HDDs/SSDs) to manage how data is organized and accessed.
- **Key Features:**
 - Supports up to 4 primary partitions, or 3 primary + 1 extended partition.
 - Maximum disk size: 2 TB
 - Stores partition table and bootloader in the first 512 bytes of the disk.
 - Commonly used in older systems and BIOS-based systems.
- **Limitations:**
 - Doesn't support disks larger than 2 TB.
 - Limited number of partitions.
 - Single point of failure (corruption of MBR can make disk unbootable).



GPT – GUID Partition Table

- GPT is the modern replacement for MBR, used in UEFI-based systems.
- **Key Features:**
 - Supports 128 partitions by default (Windows).
 - Maximum disk size: over 9 ZB (Zettabytes).
 - Stores multiple copies of partition table for redundancy.
 - Each partition has a Globally Unique Identifier (GUID).
- **Advantages:**
 - Can handle disks larger than 2 TB.
 - More secure and robust due to CRC checks and backup tables.



Basic Disks

- A Basic Disk is the default disk type in Windows operating systems.
- **Key Features:**
 - Uses MBR or GPT partition styles.
 - Can create primary partitions, extended partitions, and logical drives.
 - Supports standard partition operations (format, delete, mark active).
- **Use Case:**
 - Best for simpler systems where you don't need advanced volume management.



Dynamic Disks

- A Dynamic Disk allows more advanced volume management in Windows.
- **Key Features:**
 - Supports spanned, striped (RAID 0), mirrored (RAID 1), and RAID 5 volumes.
 - Allows volumes to span multiple physical disks.
 - Uses Logical Disk Manager (LDM).
- **Considerations:**
 - Not compatible with all OSes (e.g., some Linux distros or older Windows).
 - More complex to manage and recover if disk fails.



Storage Spaces

- Storage Spaces is a Windows feature (from Windows 8 onward) that allows combining multiple physical disks into one virtual pool.
- **Key Features:**
 - Supports Simple, Mirror, and Parity layouts.
 - Offers resiliency, flexibility, and expandability.
 - Automatically duplicates data for protection (in Mirror/Parity modes).
- **Benefits:**
 - Easy-to-manage software RAID.
 - Can expand storage without reformatting.
 - Works with both internal and external drives.



RAID (Redundant Array of Independent Disks)

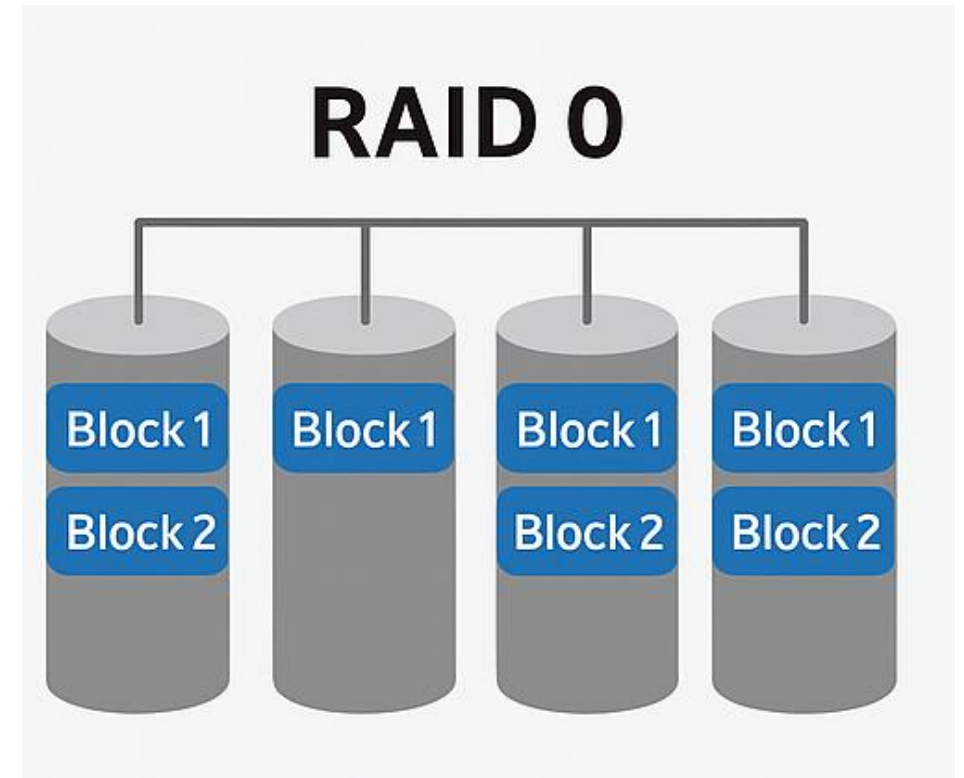
- RAID is a storage technology that combines multiple physical disks into one logical unit to improve performance, redundancy, or both.

Level	Description	Key Benefit
RAID 0	Striping (no redundancy)	High speed
RAID 1	Mirroring (duplicate data)	High availability
RAID 5	Striping with parity (min 3 disks)	Balance of speed & redundancy
RAID 6	Double parity (min 4 disks)	Tolerates 2 disk failures
RAID 10	Combines RAID 1 + 0	Performance + Redundancy



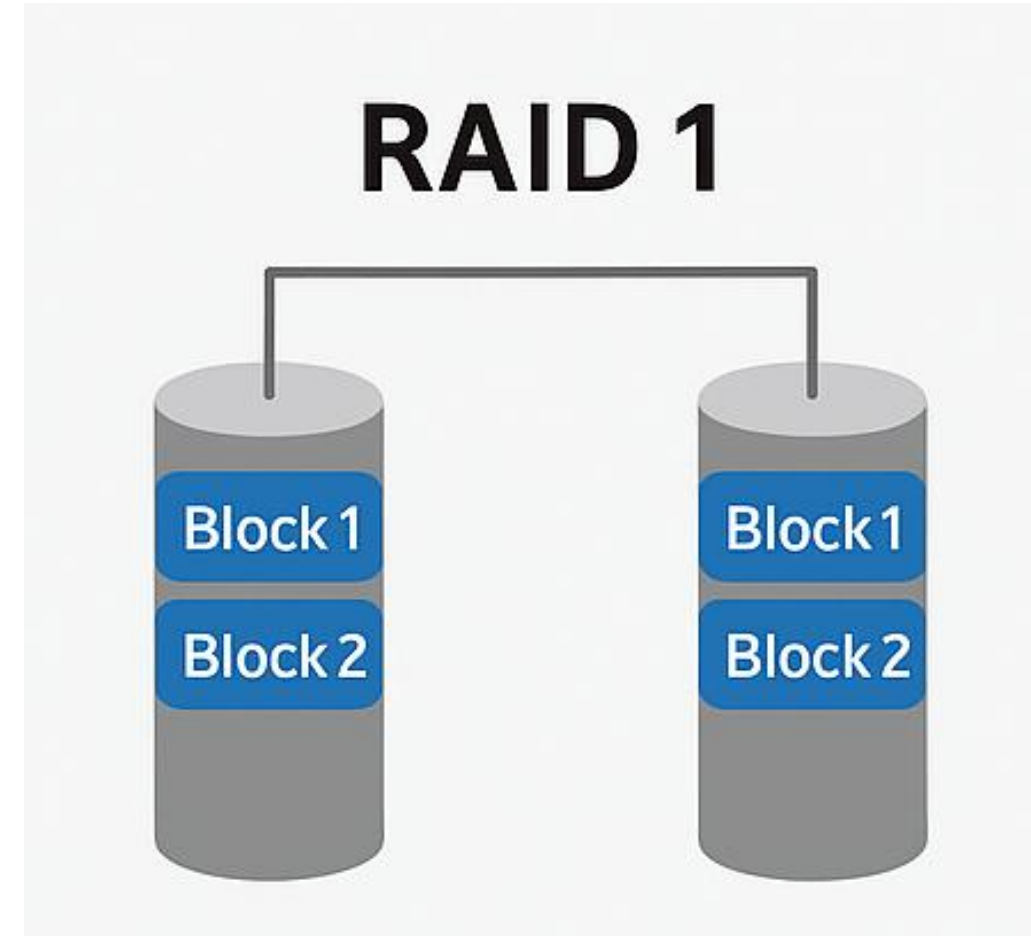
RAID 0 - Stripping

- Data is divided into blocks and spread across all disks.
- No data protection – if one disk fails, all data is lost.
- Improves performance: both read and write speeds are faster due to parallel access.
- High speed (best for large file transfers, gaming, video editing).
- Maximum use of disk space (100% of total capacity usable).
- No fault tolerance
- Not suitable for critical data



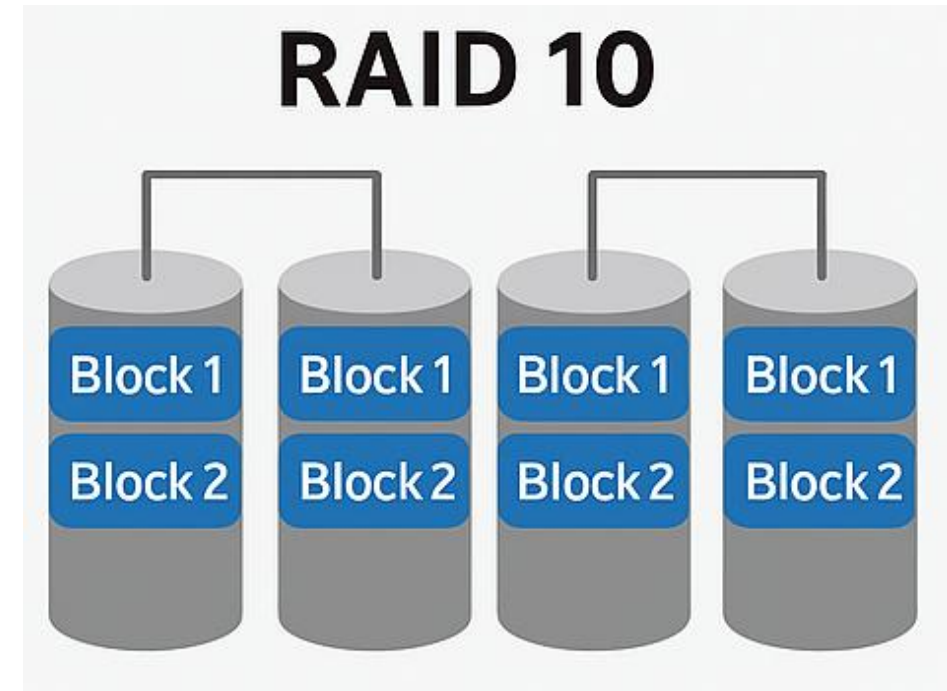
RAID 1 - Mirroring

- RAID 1 stores identical copies of data on two or more disks (mirroring).
- Key Features:
 - Full redundancy: each disk contains the same data.
 - Requires minimum 2 disks.
 - If one disk fails, the system continues to run using the mirror.
 - High data reliability
 - Simple to implement
 - Fast read performance (can read from both disks)
 - Storage efficiency = 50% (half of total space is usable)
 - Slightly slower write speed due to duplication



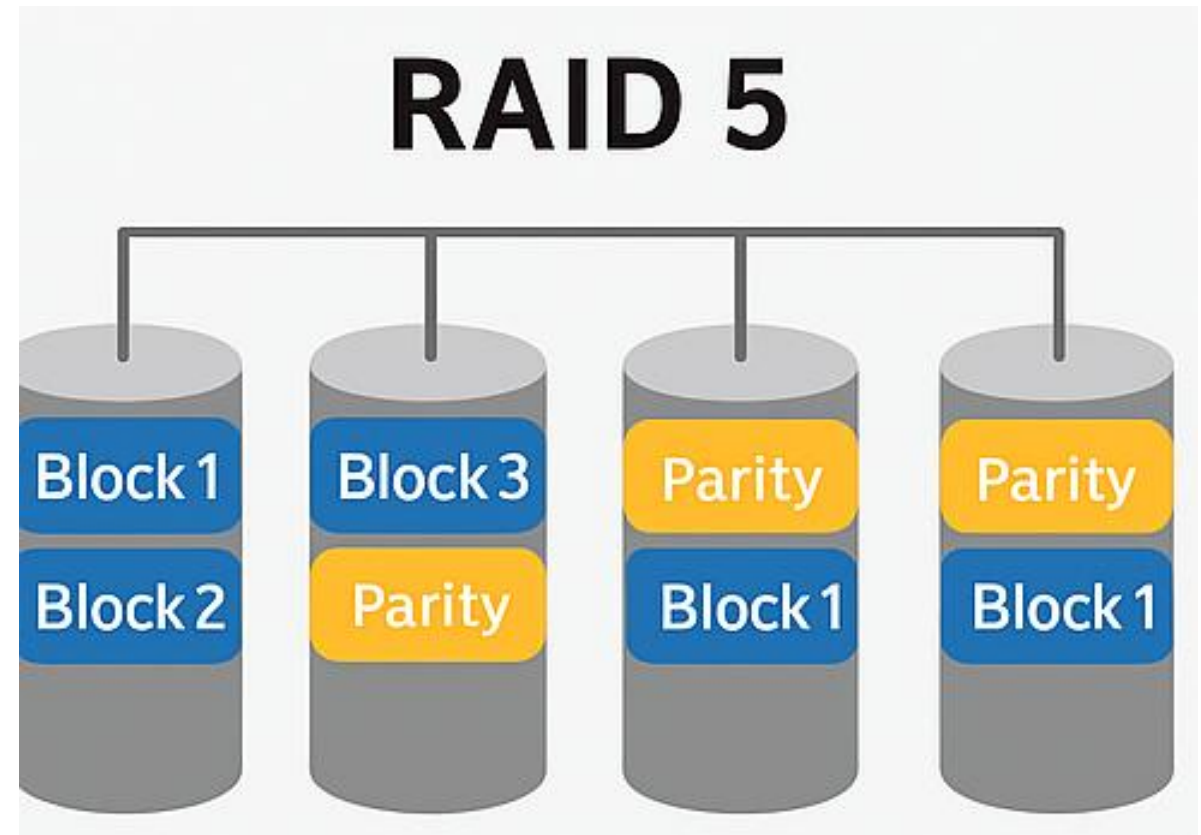
RAID 10 – Mirrored Striping (RAID 1+0)

- RAID 10 combines mirroring (RAID 1) and striping (RAID 0).
- It mirrors each striped disk pair.
- Key Features:
 - Requires at least 4 disks
 - Data is striped for speed and mirrored for redundancy
 - High performance + fault tolerance
 - Fast read and write speeds
 - Can survive multiple disk failures (if they're not in the same mirror pair)
 - Only 50% of disk space usable
 - More costly due to redundancy



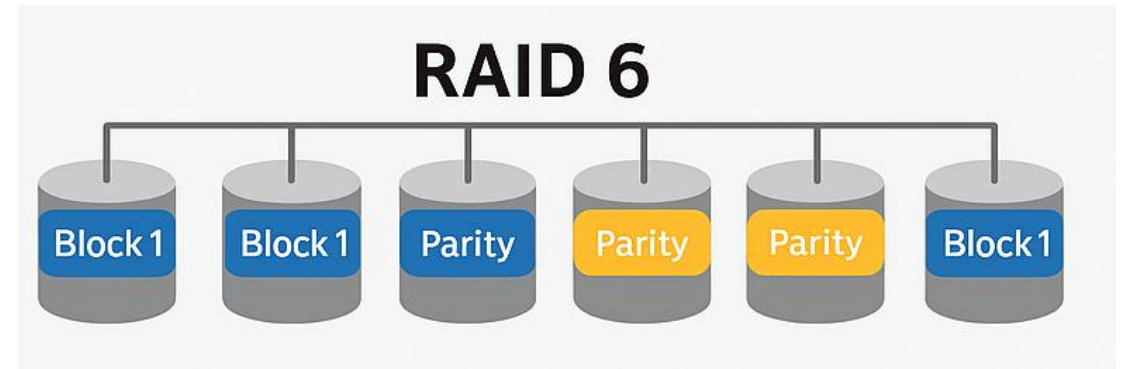
RAID 5 – Striping with Single Parity

- RAID 5 stripes data across disks and adds parity information (used for recovery) distributed across all disks.
- Key Features:
 - Requires minimum 3 disks
 - Tolerates 1 disk failure
 - Uses XOR parity to reconstruct lost data
 - Good storage efficiency (e.g., 3 disks = 66% usable)
 - Read performance is good
 - Fault tolerant
 - Write performance slower due to parity calculation
 - Rebuild time after a disk failure is long
 - Vulnerable during rebuild process



RAID 6 – Striping with Double Parity

- RAID 6 is similar to RAID 5 but uses two parity blocks. It provides higher fault tolerance.
- Key Features:
 - Requires minimum 4 disks
 - Can tolerate 2 disk failures simultaneously
 - Distributes 2 parity blocks across the array
 - High fault tolerance
 - Good read performance
 - Safer than RAID 5 during rebuild
 - Slower writes due to double parity
 - More complex
 - Storage efficiency is lower than RAID 5



RAID Summary

RAID	Min Disks	Fault Tolerance	Storage Efficiency	Read Speed	Write Speed	Use Case
1	2	1 disk	50%	High	Medium	Critical data, boot drives
10	4	1 per mirror pair	50%	Very High	High	Databases, virtualization
5	3	1 disk	$(N-1)/N$	Good	Moderate	File servers, general storage
6	4	2 disks	$(N-2)/N$	Good	Slower	Archiving, large storage arrays



iSCSI (Internet Small Computer Systems Interface)

- iSCSI is a protocol that allows block-level access to storage devices over TCP/IP networks.

- **How it Works:**

- Initiator (client): Sends I/O requests (like a disk).
- Target (server): Provides access to storage blocks.
- Appears to OS like a local disk, even though it's over a network.

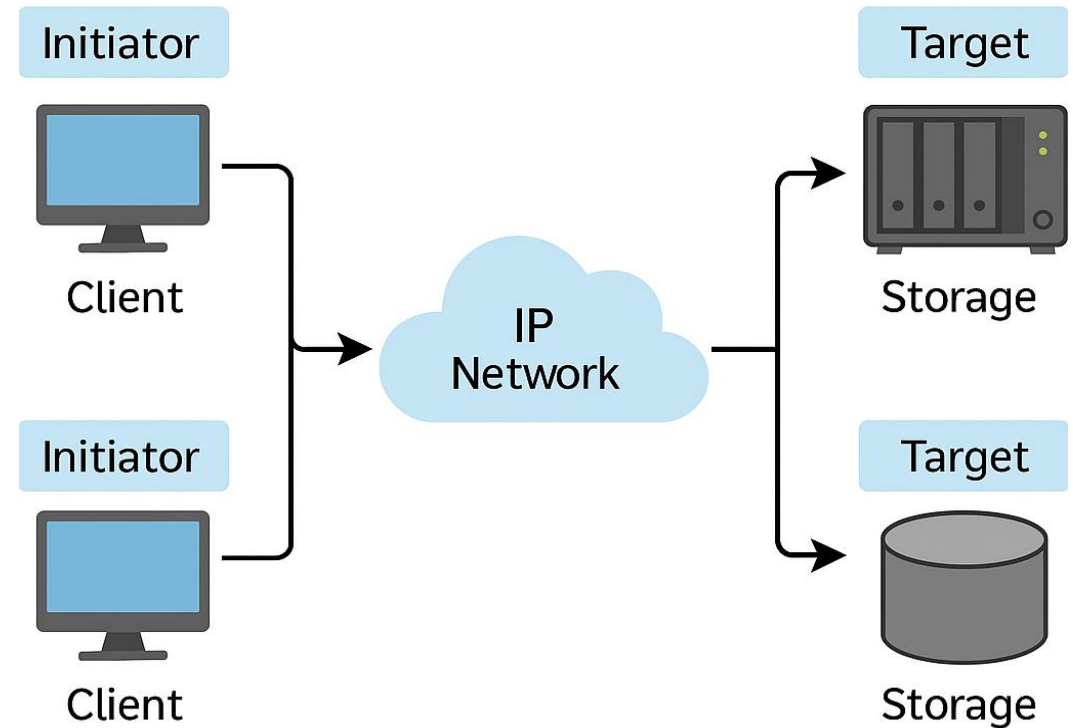
- **Use Cases:**

- SAN (Storage Area Network) environments.
- Centralized storage in virtualized and enterprise systems.

- **Benefits:**

- Low cost (uses Ethernet instead of Fibre Channel).
- Scalable and flexible storage access.

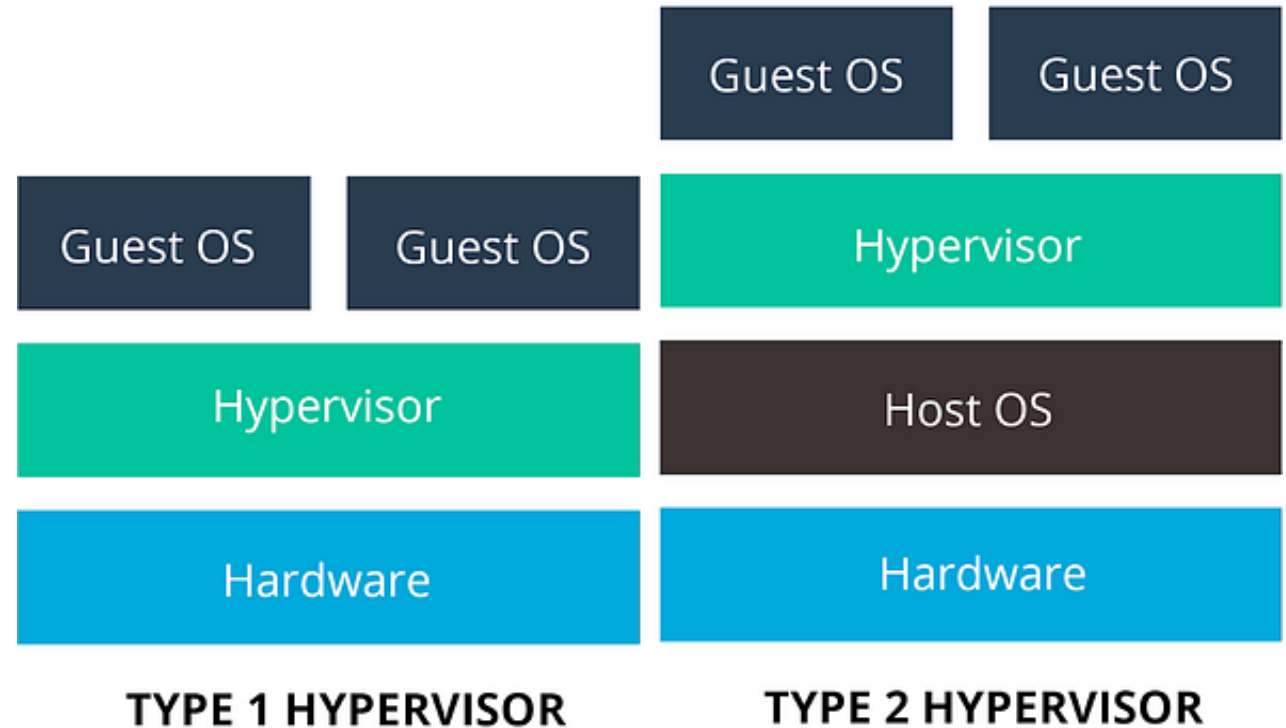
iSCSI Architecture



iSCSI Architecture

Virtualization

- Virtualization can be achieved using a tool/software called "Hypervisor".
- Hypervisors are of 2 types:
 - Type 1 Hypervisor (Bare-metal Hypervisor)
 - Type 2 Hypervisor (Guest-Based Hypervisor)



Type 1 Hypervisor

- Type 1 Hypervisor (Bare-metal Hypervisor)
- It does not require additional OS to function.
- Type 1 is mostly used in the production environments as they provide best performance.
- but they require licenses.
- ex:
 - Microsoft Hyper-V
 - VMWare ESXi
 - Citrix XenServer
 - Proxmox



Type 2 Hypervisor

- Operating system is needed. without OS, this HV will not work.
- Useful for testing/home purpose.
- Their performance is not equivalent to Type 1.
- Ex:
 - Microsoft Hyper-V (Windows 10/11 - Enterprise/Pro)
 - VMWare workstation (free)
 - Oracle Virtual Box (free)
 - VMWare player (free)
 - VMWare fusion (Paid - MacOS)
 - Linux KVM (Kernel-based Virtual Machine) (free)







That's all Folks!